Basic Statistics

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Preface

1 Introduction to Statistics

- **Data:** Data consists of information coming from observations, counts, measurements, or responses. *(Ron Larson)*
- **Statistics:** Statistics is the science of collecting, organizing, analyzing, and interpreting data in order to make decisions. *(Ron Larson)*

1.1 Data Sets

There are two types of data sets you will use when studying statistics: **populations** and **samples**.

- **Population:** The entire collection of objects or outcomes about which data are collected.
  - A **parameter** refers to a numerical quantity computed from a population.
- **Sample:** A subset of the population containing the observed objects or outcomes and the resulting data.
  - A **statistic** (rather than the field of "statistics") refers to a numerical quantity computed from sample data (e.g., the mean, the median, the maximum).

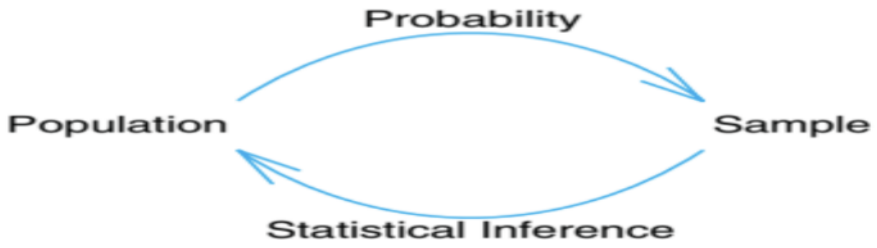
1.2 Branches of Statistics

The study of statistics has two major branches:

1. **Descriptive Statistics:** Describes the important characteristics/properties of the data using measures of central tendency (mean, median, mode) and measures of dispersion (range, standard deviation, variance). Data can be summarized and represented using charts, tables, and graphs.
2. **Inferential Statistics:** Uses data from a sample to make inferences about the larger population from which the sample was drawn. The goal is to draw conclusions from a sample and generalize them to the population using probability theory.

   Common methodologies include hypothesis tests and Analysis of Variance (ANOVA).

Figure 1 shows the fundamental relationship between probability and inferential statistics:



1.3 Application of Probability and Statistics in Engineering

- **Electrical Engineering:** Signals and noise are analyzed using probability theory.
- **Civil, Mechanical, & Industrial Engineering:** Used to test and account for variations in materials and goods.
- **Chemical Engineering:** Used to assess experimental data and control/improve chemical processes.

1.3.1 Why Textile Engineers Study Statistics

Textile engineers study variations occurring from material to material, test to test, sample to sample, machine to machine, time to time, and place to place. Key questions include:

1. How many tests are to be carried out to get the desired results?
2. How to analyze the collected data?
3. How to interpret the results of the analysis?

1.4 Types of Variables / Data

Data sets can consist of two types of data: **Qualitative (Categorical)** and **Quantitative**.

1.5 Qualitative (Categorical) Variables

Consist of attributes, labels, or non-numerical entries. Values are intrinsically non-numeric. * *Examples:* Cause of death, nationality, race, gender, severity of pain (mild, moderate, severe).

1.6 Quantitative Variables

Consist of numerical measurements or counts where arithmetic operations make sense. * *Examples:* Number of defective items, systolic blood pressure, height, age.

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1.6.1 Sub-divisions

- **Discrete Variables:** Have a set of possible values that is finite or countably infinite, with gaps between possible values. Often integer values/counts. *(e.g., number of pregnancies, shoe size)*
- **Continuous Variables:** Can take any value within an interval of the real line with no gaps. *(e.g., duration of a seizure, BMI, height)*

1.7 Data Collection and Experimental Design

Guidelines for Designing a Statistical Study

1. Identify the variable(s) of interest and the target population.
2. Develop a detailed plan for collecting data (ensure samples are representative).
3. Collect the data.
4. Describe the data using descriptive statistics.
5. Interpret data and make decisions using inferential statistics.
6. Identify possible errors.

1.8 Observational Study vs. Experiment

- **Observational Study:** The researcher observes and measures characteristics of interest without influencing or changing existing conditions.
- **Experiment:** A treatment is deliberately applied to part of a population (treatment group) and responses are observed. A control group (often given a placebo) is used for comparison.

1.9 Data Management & Arrays

Data management refers to processing steps occurring after collection but before analysis. Statistical software operates primarily on rectangular arrays where **rows = cases** and **columns = variables**.

Example: Medical Expenditures Data Set

Table 1.1: Typical Rectangular Data Set

ID	Age	State	Expenditures
132414	33	DHK	375
454543	25	CTG	1450
232134	41	RAJ	908
314132	55	KHL	3480
453255	46	CTG	2480
543545	35	CTG	8550
342342	28	BAR	320

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ID	Age	State	Expenditures
563654	41	NOA	24350
465566	27	DHK	1850

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2 Frequency distribution

1) Ungrouped (discrete) - Usually used to understand the distribution of count data

2) Grouped (continuous) - Usually used to understand the distribution of scale data

Example-1: Construct the frequency distribution for the following data of number of weaving defects observed in the 30 pieces of fabric. 4, 0, 1, 5, 2, 6, 0, 4, 4, 1, 5, 4, 5, 4, 3, 0, 2, 3, 3, 2, 6, 4, 1, 2, 1, 3, 4, 3, 2, 3.

Solution: Here, Minimum= 0 Maximum=6

So, the frequency table/distribution of number of weaving defects is:

Number of defects	Tally	Frequency
0		
1		
2		
3		
4		
5		
6		

Answer:

No. of defects	Freq
0	3
1	4
2	5
3	6
4	7
5	3
6	2

Example-2: Following are the results of 'hairiness index' obtained from 20 tests:

6.72, 6.52, 6.00, 6.85, 6.08, 6.64, 6.88, 6.91, 6.35, 6.52, 6.26, 7.20, 7.12, 6.36, 6.82, 6.53, 6.78, 6.75, 6.49, 6.62.

Construct the grouped frequency distribution by taking suitable/ appropriate class interval in exclusive method.

Solution:

Here, $n = 20$

$Minimum = 6.0$; $Maximum = 7.2$;

Range, $R = Maximum - Minimum = 7.20 - 6.00 = 1.20$.

- Number of classes ,

$$k = \sqrt{n} = \sqrt{20} = 4.47 \approx 5$$

- So, class interval/ width ,

$$w = \frac{R}{k} = \frac{1.2}{5} = 0.24 \approx 0.25$$

Hence in exclusive method; required class limits are: [6.0,6.25), [6.25, 6.50), [6.50,6.75), [6.75,7.00), [7.00,7.25).

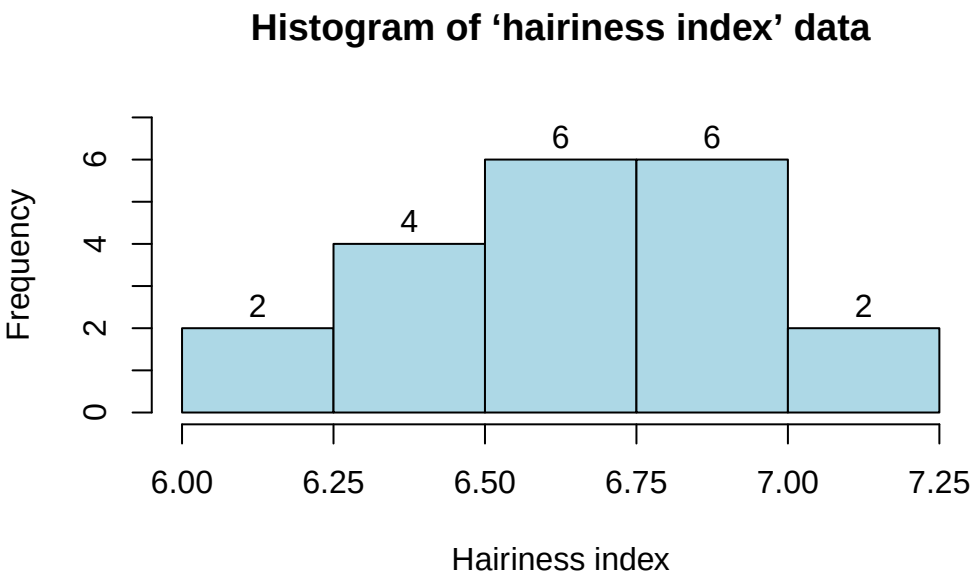
Frequency distribution of hairiness index (n=20):

Class interval	Tally	Frequency
[6.00,6.25)		
[6.25, 6.50)		
[6.50,6.75)		
[6.75,7.00)		
[7.00,7.25)		
Total		

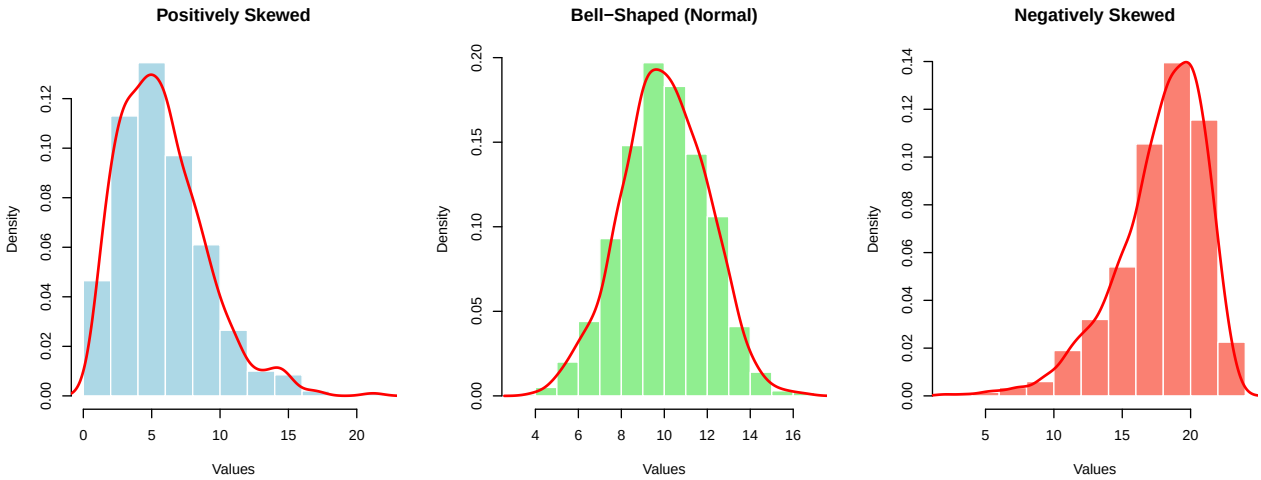
Ans.

Class interval	Frequency
[6.00,6.25)	2
[6.25, 6.50)	4
[6.50,6.75)	6
[6.75,7.00)	6
[7.00,7.25)	2
Total	$n = 20$

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2.1 Histogram and shape of distribution



Relative, percent and cumulative frequency distribution- example with the ‘hairiness index’ data

Class interval	Frequency	Relative frequency (rf)	Percent frequency(pf)	Cumulative frequency (cf)
[6.00,6.25)	2	0.10	10	2
[6.25,6.50)	4	0.20	20	6
[6.50,6.75)	6	0.30	30	12
[6.75,7.00)	6	0.30	30	18
[7.00,7.25)	2	0.10	10	20
Total	**20**	**1.00**	**100.00**	

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2.2 Exercise

1. **Construct** the frequency distribution for the following data of number of weaving defects observed in the 30 pieces of fabric. 4, 0, 1, 5, 2, 6, 0, 4, 4, 1, 5, 4, 5, 4, 3, 0, 2, 3, 3, 2, 6, 4, 1, 2, 1, 3, 4, 3, 2, 3

2. **Construct** the *ungrouped* frequency distribution for the following data of number of end breaks observed on a ring frame for the 30 days.

   10, 8, 5, 10, 7, 7, 11, 8, 10, 12, 11, 12, 6, 6, 7, 10, 10, 12, 10, 10, 6, 7, 8, 10, 8, 8, 12, 11, 10, 10.

3. Following are the figures of number of workers absent in a textile mill every day:

   2, 3, 1, 1, 2, 3, 2, 0, 0, 1, 2, 5, 0, 0, 2,1, 1, 4, 3, 2, 0, 2, 1, 2, 2, 4, 3, 3, 3, 4.

   Construct the ungrouped frequency distribution for the above data.

**4. Following are the results of 'hairiness index' obtained from 20 tests:

6.72, 6.52, 6.00, 6.85, 6.08, 6.64, 6.88, 6.91, 6.35, 6.52, 6.26, 7.20, 7.12, 6.36, 6.82, 6.53, 6.78, 6.75, 6.49, 6.62.

Construct the grouped frequency distribution by taking classes 6.00–6.20, 6.20–6.40 and so on.

5. Following are the number of accidents per day observed in a textile mill during a month of 30 days:

   4, 3, 3, 3, 4, 2, 2, 1, 2, 0, 2, 3, 4, 1, 1, 2, 0, 0, 5, 2, 1, 0, 0, 2, 3, 2, 1, 1, 3, 2.

   Construct an ungrouped frequency distribution.

6. Thirty pieces of fabric were observed for the number of defects and following results were obtained. 1, 0, 0, 2, 3, 2, 1, 1, 3, 2, 4, 3, 4, 1, 1, 2, 0, 0, 5, 2, 3, 3, 4, 2, 2, 1, 2, 0, 2, 3 Construct an ungrouped frequency distribution.

7. **Construct** the frequency distribution for the following results of the daily sale (in 1000 Tk.) in a textile firm recorded for a month.

   21, 21, 21, 22, 26, 20, 20, 20, 21, 21, 20, 20, 22, 20, 21, 23, 21, 22, 21, 25, 18, 24, 25, 19, 25, 18, 26, 23, 18, 24.

   Also **comment** about shape of the distribution of daily sales.

**8. Thirty linear density tests made on the yarn have shown following results.

14.8, 14.2, 13.8, 13.5, 14.0, 14.2, 14.3, 14.6, 13.9, 14.0, 14.1, 13.2, 13.0, 14.2, 13.5, 13.0, 12.8, 13.9, 14.8, 15.0, 12.8, 13.4, 13.2, 14.0, 13.8, 13.9, 14.0, 14.0, 13.9, 14.8.

Construct the grouped frequency distribution by taking class intervals 12.6–13.0; 13.1–13.5 and so on.

9. Following is the distribution of fabric samples according to the strength of fabric:

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Strength	150–155	155–160	160–165	165–170	170–175
No. of samples	8	12	17	10	7

Draw the histogram for the above data and interpret it.

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3 Introduction to Central Tendency

Central tendency is a statistical concept used to describe the **center** or **typical value** of a dataset.

Instead of looking at every individual data point, measures of central tendency provide a single summary value that represents the entire distribution.

The three most common measures are the **mean, median**, and **mode**, each offering a different perspective on what constitutes the "center" of the data depending on its distribution and the presence of outliers.

3.1 Common Measures of Central Tendency

1) Mean / Average / Arithmetic mean
2) Median
3) Mode

Case I: Formula for Ungrouped / Raw Data

Consider a variable X with "n" observations that is:

$$X = \{x_1, x_2, \dots, x_n\}$$

A. Mean of X:

$$\bar{X} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum x}{n}$$

B. Median: Middle most observation of an arranged data.

C. Mode: Most frequent observation(s) in a data.

Example-1

Ten-count tests made on certain yarn have shown the following results: 22.8, 23.2, 22.9, 22.6, 23.4, 23.0, 23.1, 23.0, 22.9, 23.0

Compute mean, median and mode of the above data:

Solution:

$$n = 10; \sum x = 22.8 + 23.2 + \cdots + 23.0 = 229.9$$

A) Mean:

$$\text{Mean} = \frac{\sum x}{n} = \frac{229.9}{10} = 22.99$$

B) Median: Arranged X: 22.6, 22.8, 22.9, 22.9, 23.0, 23.0, 23.0, 23.1, 23.2, 23.4

$$\text{Median} = \frac{23.0 + 23.0}{2} = 23.0$$

C) Mode:

$$\text{Mode} = 23.0 \text{ (observed 3 times uniquely)}$$

Case II: Formula for Grouped / Frequency Distribution

Consider the following frequency distribution:

Class interval	Mid value (x_i)	Frequency (f_i)
$[L_1, L_2)$	$x_1 = (L_1 + L_2)/2$	f_1
$[L_2, L_3)$	x_2	f_2
…	…	…
$[L_k, L_{k+1})$	x_k	f_k
Total		$N = \sum_{i=1}^{k} f_i$

A) Mean:

$$\text{Mean} = \frac{f_1x_1 + f_2x_2 + \cdots + f_kx_k}{f_1 + f_2 + \cdots + f_k} = \frac{\sum_{i=1}^{k} f_ix_i}{N} = \frac{\sum fx}{N}$$

B) Median:

$$\text{Median} = L_o + \frac{c}{f}\left(\frac{N}{2} - p.cf\right)$$

Where:

L_o = Lower limit of Median class

c = class interval of Median class

f = frequency of Median class

$p.cf$ = cf (cumulative frequency of pre median class)

C) Mode: [For approximately symmetric distribution]

$$\text{Mode} = L_o + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times c$$

Where:

L_o = Lower limit of Modal class

c = class interval of Modal class

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Δ_1 = differences between frequencies of pre modal and modal class

Δ_2 = differences between frequencies of post modal and modal class

Example-2 Determine the mean, median and mode from the following frequency distribution and comment on it.

Length of fiber(mm)	10–15	15–20	20–25	25–30	30–35	35–40
No. of fibers	12	19	26	21	15	9

Solution:

Length of fiber(mm)	f	Mid-value (x)	fx	cf
10–15	12	12.5	150	12
15–20	19	17.5	332.5	31
20–25	26	22.5	585	57
25–30	21	27.5	577.5	78
30–35	15	32.5	487.5	93
35–40	9	37.5	337.5	102
Total	$N = 102$		$\sum fx = 2470$	

A) Mean:

$$\text{Mean} = \frac{\sum fx}{N} = \frac{2470}{102} = 24.22$$

B) Median: To determine median we have to identify the "Median" class. First calculate cumulative frequency (cf). Then, calculate:

$$\frac{N}{2} = \frac{102}{2} = 51$$

This is less than the $cf = 57$ which corresponds to class interval 20-25.

$$\text{Median} = L_o + \frac{c}{f}\left(\frac{N}{2} - p.cf\right) = 20 + \frac{5}{26}(51 - 31) = 23.85$$

C) Mode: Here the highest frequency is $f = 26$, located in the class interval 20-25. So, 20-25 is the modal class.

$$\Delta_1 = 26 - 19 = 7; \quad \Delta_2 = 26 - 21 = 5$$

$$\text{Mode} = L_o + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times c = 20 + \frac{7}{7+5} \times 5 = 22.92$$

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4 Partition values (quantiles)

Common partition values are:

1. **Percentiles**
2. **Quartiles**
3. **Deciles**

4.1 Percentiles

Percentiles are 99 values which divide a data set into approximately 100 equal parts. There are 99 percentiles, normally denoted as:

$$P_1, P_2, \ldots, P_{25}, \ldots, P_{50}, \ldots, P_{75}, \ldots, P_{99}$$

4.1.1 A) Formula for Computing j^{th} Percentile from Raw / Ungrouped Data

$$P_j = j^{\text{th}} \text{ Percentile} = \left[j \times \frac{(n+1)}{100} \right]^{\text{th}} \text{ value}$$

Example 4.1. Ten-count tests made on a certain yarn have shown the following results:
`22.8, 23.2, 22.9, 22.6, 23.4, 23.0, 23.1, 23.0, 22.9, 23.0`

Compute P_{25} and P_{50}.

4.1.1.1 Solution

Here, $n = 10$.

Arranged data:
`22.6, 22.8, 22.9, 22.9, 23.0, 23.0, 23.0, 23.1, 23.2, 23.4`

Computing P_{25}:

$$\begin{aligned} P_{25} &= \left[25 \times \frac{(10+1)}{100} \right]^{\text{th}} \text{ value} = 2.75^{\text{th}} \text{ value} \\ &= 2^{\text{nd}} \text{ value} + 0.75 \times (3^{\text{rd}} \text{ value} - 2^{\text{nd}} \text{ value}) \\ &= 22.8 + 0.75 \times (22.9 - 22.8) = 22.875 \end{aligned}$$

Computing P_{50}:

$$P_{50} = \left[50 \times \frac{(10+1)}{100}\right]^{\text{th}} \text{ value} = 5.5^{\text{th}} \text{ value}$$
$$= 5^{\text{th}} \text{ value} + 0.5 \times (6^{\text{th}} \text{ value} - 5^{\text{th}} \text{ value})$$
$$= 23.0 + 0.5 \times (23.0 - 23.0) = 23.0$$

4.1.2 B) Formula for Computing j^{th} Percentile from Grouped Data / Frequency Distribution

$$P_j = L_0 + \frac{c}{f}\left(\frac{N}{100} \times j - \text{p.cf}\right)$$

Where:

- L_0 = Lower limit of the jth percentile class
- c = Class interval of the jth percentile class
- f = Frequency of the jth percentile class
- p.cf = Cumulative frequency of the class preceding the percentile class

Example 4.2. The following data are related to the linear density of yarn:

Linear density	13.00–13.25	13.25–13.50	13.50–13.75	13.75–14.00	14.00–14.25	14.25–14.50	14.50–14.75
No. of tests	8	12	20	25	22	10	4

Compute and explain P_{50} and P_{75}.

4.1.2.1 Solution

First, we compute the cumulative frequency (cf):

Linear density	Frequency (f)	Cumulative Frequency (cf)
13.00–13.25	8	8
13.25–13.50	12	20
13.50–13.75	20	40
13.75–14.00	**25**	**65**
14.00–14.25	22	87
14.25–14.50	10	97
14.50–14.75	4	101
Total	$N = 101$	

Finding P_{50}:

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To locate the 50th percentile class, compute:

$$\frac{N}{100} \times 50 = \frac{101}{100} \times 50 = 50.5$$

The cumulative frequency just greater than 50.5 is 65.
So, **13.75–14.00** is the 50th percentile class.

$$\begin{aligned} P_{50} &= L_0 + \frac{c}{f}\left(\frac{N}{100} \times 50 - \text{p.cf}\right) \\ &= 13.75 + \frac{0.25}{25} \times (50.5 - 40) \\ &= 13.855 \end{aligned}$$

Interpretation:
This means 50% of the 101 observations (i.e., $50\% \times 101 = 50.5 \approx 51$ observations) have a linear density lying at or below 13.855.

(Try computing P_{75} yourself as practice.)

4.2 Quartiles

Quartiles are 3 values that divide a data set into approximately 4 equal parts:

1. Q_1 (1st quartile / lower quartile)
2. Q_2 (2nd quartile / median)
3. Q_3 (3rd quartile / upper quartile)

Note that:

- $Q_1 = P_{25}$
- $Q_2 = P_{50} = \text{Median}$
- $Q_3 = P_{75}$

4.2.1 Formula for Raw Data

$$Q_j = \left(j \times \frac{n+1}{4}\right)^{\text{th}} \text{ value}, \quad \text{for } j = 1, 2, 3$$

Example 4.3. Find Q_1, Q_2, and Q_3 for the following data representing CPU time of 11 jobs (in seconds):
`59, 139, 46, 37, 42, 30, 55, 56, 36, 82, 29`

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4.2.1.1 Solution

Here, $n = 11$.

Sorted data:
`29, 30, 36, 37, 42, 46, 55, 56, 59, 82, 139`

$$Q_1 = P_{25} = \left(1 \times \frac{11+1}{4}\right)^{\text{th}} \text{ value} = 3^{\text{rd}} \text{ value} = 36$$
$$Q_2 = P_{50} = \left(2 \times \frac{11+1}{4}\right)^{\text{th}} \text{ value} = 6^{\text{th}} \text{ value} = 46$$
$$Q_3 = P_{75} = \left(3 \times \frac{11+1}{4}\right)^{\text{th}} \text{ value} = 9^{\text{th}} \text{ value} = 59$$

4.2.2 Interquartile Range (IQR)

The **Interquartile Range** is the difference between the third quartile (Q_3) and the first quartile (Q_1):

$$\text{IQR} = Q_3 - Q_1$$

Exercise 4.1 (Question on Interpretation)**.** In a class of 40 students, mid-term test marks were collected and P_{20} was found to be 23.

1. What does $P_{20} = 23$ mean?
2. Approximately how many students received a score of 23 or less?

Answer:
$P_{20} = 23$ means that 20% of the students scored 23 or fewer marks.
Number of students scoring 23 or less $= 40 \times 20\% = 8$ students.

4.3 Box-Plots and the Five-Number Summary

Five-Number Summary

The **minimum**, **first quartile** (Q_1), **median** (Q_2), **third quartile** (Q_3), and **maximum**, taken together, are referred to as the **five-number summary** of a data set.

4.3.1 Box-and-Whisker Plot

A **box-plot** (or box-and-whisker plot) is a chart used in exploratory data analysis to visually display the distribution of numerical data and its skewness through the five-number summary.

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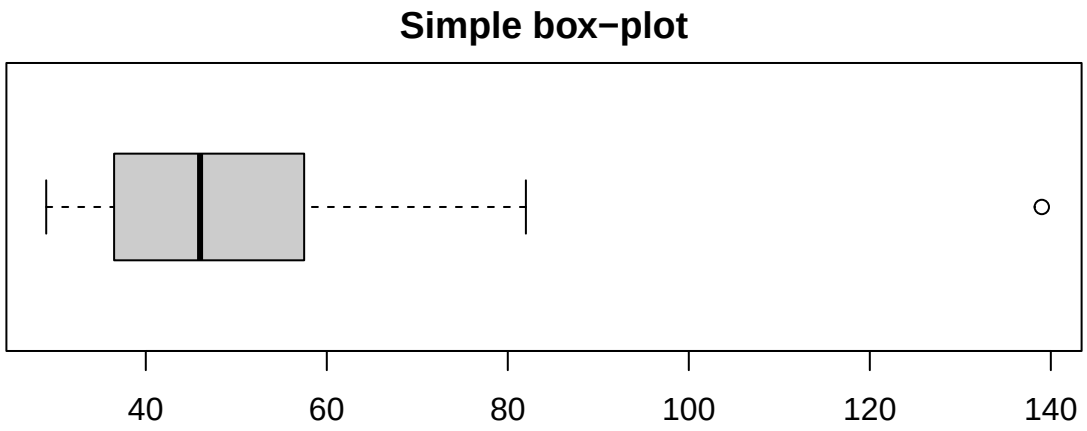


Figure 4.1: Simple Box-plot of CPU Time (in seconds)

4.3.2 Significance of a Box-Plot

- Provides a quick visual summary of data distribution.
- Shows the central tendency from the line inside the box (Median).
- Displays total spread (Minimum to Maximum) as well as middle 50% spread (IQR $= Q_3 - Q_1$).
- Gives clear indications about the skewness/shape of the distribution.

4.3.3 Parallel / Comparative Box-Plots

Box-plots are frequently used to compare multiple groups or populations side-by-side using the same scale.

4.3.4 Example: Tensile Strength of Paper

(Adapted from D.C. Montgomery, Applied Statistics and Probability for Engineers)

A paper manufacturer wants to improve product tensile strength. Engineering tests four levels of hardwood concentration in paper pulp: 5%, 10%, 15%, and 20%. Six test specimens are prepared at each concentration level.

4.3.4.1 Data Table

Hardwood Conc. (%)	Obs 1	Obs 2	Obs 3	Obs 4	Obs 5	Obs 6	Totals	Averages
5%	7	8	15	11	9	10	60	10.00
10%	12	17	13	18	19	15	94	15.67
15%	14	18	19	17	16	18	102	17.00
20%	19	25	22	23	18	20	127	21.17
Overall							**383**	**15.96**

4.3.4.2 Interpretation

- **Effect on Strength:** Increasing hardwood concentration consistently increases mean and median tensile strength.
- **Symmetry & Variability:** Distribution across levels is relatively symmetric, and disper-

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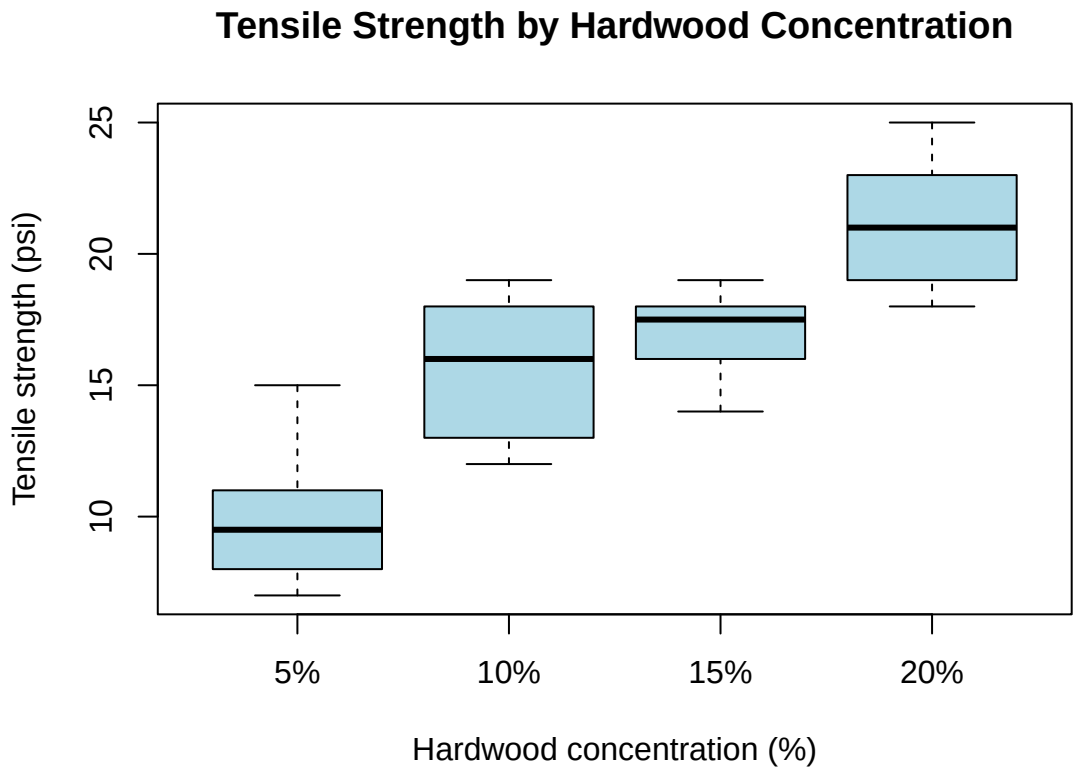


Figure 4.2: Box plots of hardwood concentration data (Montgomery, Table 13.1)

4.4 Exercises

1. **Daily Wages:** Find the 3rd quartile and 60th percentile for the following distribution of daily wages of temporary workers, and comment on the results:

Daily wages (Rs)	40–49	50–59	60–69	70–79	80–89
No. of workers	15	20	30	45	25

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2. **Fabric Strength:** Calculate the 3rd decile (D_3), 2nd percentile (P_2), and 70th percentile (P_{70}):

Fabric strength	45–49	50–54	55–59	60–64	65–69	70–74
No. of samples	5	10	15	20	10	5

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3. **Coefficient of Variation (CV%):** Calculate Q_1 and P_{70} for the following data and comment on them:

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CV%	3.0–3.5	3.5–4.0	4.0–4.5	4.5–5.0	5.0–5.5
No. of tests	25	4	10	16	10

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4. **Goods Weight:** Compute the Median and 38th percentile (P_{38}) for the following data:

Weight (kg)	10–19	20–29	30–39	40–49	50–59	60–69
No. of goods	6	10	16	14	8	4

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5. **Yarn Unevenness (U%):** Compute the 1st quartile (Q_1), 7th decile (D_7), and 38th percentile (P_{38}) for the following raw measurements:
`2.25, 4.25, 2.65, 3.85, 5.82, 3.42, 4.44, 2.86, 2.88, 4.46`

---

6. **Loom Comparison:** A textile company measures fabric strength (psi) across 2 randomly selected looms:

Loom	Measurements
A	99, 97, 97, 96, 97, 96, 92, 98
B	94, 95, 90, 92, 93, 94, 90, 92

i. Construct a parallel/comparative box-plot comparing fabric strength for Loom A and Loom B.
ii. Based on the box-plot, which loom produces better fabric in terms of strength? Which loom exhibits lower variability?

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5 Measures of variation

5.1 Chapter Overview & Learning Objectives

By the end of this chapter, learners will be able to:

- Explain why central tendency alone fails to characterize quality variation in fiber, yarn, and fabric manufacturing.
- Calculate and interpret absolute and relative measures of dispersion for both ungrouped lab trials and grouped production data.
- Apply industrial benchmarks (such as Uster Statistics) to evaluate process capability using Coefficient of Variation ($CV\%$).

5.2 Variation or dispersion

Variability in data means lack of uniformity. It is also referred to as spread, scatter, or dispersion. We turn now to a discussion of some commonly used measures of variability.

5.3 Measures of variation for ungrouped or raw data

5.3.1 Range

Range, $R = Maximum\ \ value - Minimum\ \ value$

- Heavily influenced by extreme values

5.3.2 Inter-quartile-range

Inter-quartile-range, $IQR = Q_3 - Q_1$

- It exhibits the variability in the middle 50% of the observations.
- The interquartile range is less sensitive to the extreme values in the sample than is the ordinary sample range.

5.3.3 Variance and Standard deviation

- Sample variance (denoted by English Letter s^2),

$$s^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + ... + (x_n - \bar{x})^2}{n-1} = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

$$\therefore \;\; s^2 = \frac{\sum x_i^2 - n(\bar{x})^2}{n-1} \tag{5.1}$$

and Sample SD,

$$s = \sqrt{s^2}$$

- Population variance (denoted by Greek Letter σ^2),

$$\sigma^2 = \frac{(x_1 - \mu)^2 + (x_2 - \mu)^2 + ... + (x_N - \mu)^2}{N} = \frac{\sum(x_i - \mu)^2}{N} = \frac{\sum x_i^2 - N(\mu)^2}{N}$$

$$\therefore \;\; \sigma^2 = \frac{\sum x^2}{N} - \mu^2 \tag{5.2}$$

and Population SD is

$$\sigma = \sqrt{\sigma^2}$$

Note: In modern statistics and quality management, **consistency is the hallmark of quality.** While the **Mean** tells you if you are hitting your target, the **Variation (Standard Deviation)** tells you how reliable you are. High variation is often referred to as the "enemy" of quality because it creates unpredictability.

Problem : Ten-count tests made on certain yarn have shown following results.

22.8, 23.2, 22.9, 22.6, 23.4, 23.0, 23.1, 23.0, 22.9, 23.0

Compute sample mean, sample variance and sample standard deviation.

Solution: Suppose X be the count test.

Here, $n = 10$; $\sum x = 229.9$,

$\sum x^2 = x_1^2 + x_2^2 + ... + x_n^2 = 22.8^2 + 23.2^2 + ... + 23.0^2 = 5285.83$

So, the **sample mean** $\bar{x} = \frac{\sum x}{n} = \frac{229.9}{10} = 22.99$.

Sample variance, $s^2 = \frac{\sum x^2 - n\cdot(\bar{x}^2)}{n-1} = \frac{5285.83 - 10\cdot(22.99^2)}{10-1} = 0.04766667$ and

Sample SD, $s = \sqrt{s^2} = \sqrt{0.04766667} = 0.218327 \approx 0.2183$.

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5.3.4 Coefficient of variation

$$CV = \frac{SD}{Mean} \tag{5.3}$$

- CV is unit free.
- A lower CV indicates that the variation is small compared to the size of the mean, which refers **greater consistency** or **reliability.**
- In general industry Standards for "Good Consistency", **CV should be less than 10%** and $CV \geq 30\%$ indicates "Poor consistency" (**vanBelle2002?**) .

Problem: **Compute** coefficient of variation of yarn-count data mentioned in Problem…..

Problem Five strength tests each carried out on the two fabrics have shown following results:

A: 15.2, 15.5, 15.0, 15.4, 15.6

B: 15.0, 14.8, 13.6, 15.5, 15.2

Find CV% for both and **decide** which fabric is more consistent.

Solution:

Problem Following data represents number of end breaks recorded on two ring frames at ten occasions.

R/F-I: 12, 10, 8, 15, 9, 11, 14, 15, 8, 10

R/F-II: 10, 11, 12, 9, 12, 9, 11, 12, 11, 10

Find which ring frame is more consistent in terms of end breaks.

Solution:

Exercise 1.1

1) An article describes an experiment to test the yield strength of circular tubes with caps welded to the ends. The first 8 yields (in kN) are: 104, 108, 126,126, 128, 128, 140, 156
Compute the sample mean, the standard deviation, and the coefficient of variation of the yield strength.
2) The "cold start ignition time" of an automobile engine is being investigated by a gasoline manufacturer. The following times (in seconds) were obtained for a test vehicle: 1.75, 1.92, 2.62, 2.35, 3.09, 3.15, 2.53, 1.91.
Calculate the sample mean, sample standard deviation and coefficient of variation of cold start ignition time.
3) Following are the results of count tests made on the production of two shifts:
Shift I: 11.7, 12.2, 12.0, 11.6, 10.9, 13.0, 11.0, 12.8, 11.5, 12.0
Shift II: 12.2, 11.6, 12.0, 12.1, 12.0, 12.3, 11.8, 11.6, 11.8, 12.0
Using CV% **decide** which production is more consistent.

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5.4 Measures of variation for grouped or raw data

Sample Variance:

$$s^2 = \frac{\sum_{i=1}^{k} f_i(m_i - \bar{x})^2}{n-1}$$

Sample Standard Deviation:

$$s = \sqrt{\frac{\sum_{i=1}^{k} f_i(m_i - \bar{x})^2}{n-1}} = \sqrt{\frac{\sum_{i}^{k} f_i m_i^2 - n \times \bar{x}^2}{n-1}}$$

where:

k = number of class intervals

f_i = frequency of the i-th class

m_i = midpoint of the i-th class interval

$n = \sum_{i=1}^{k} f_i$ = total sample size

$\bar{x} = \frac{\sum_{i=1}^{k} f_i m_i}{n}$ = grouped sample mean

Example: Dataset: Yarn Breaking Force (cN)

Force (cN)	Frequency (f)
240-250	8
250-260	22
260-270	40
270-280	20
280-290	10

a) **Compute** sample standard deviation and CV%.

b) Does the quality of yarn acceptable? Why or why not?

Solution:

Force (cN)	Mid value (m_i)	Frequency (f_i)	$f_i m_i$	$f_i m_i^2$
240-250	245	8		
250-260	255	22		
260-270	265	40		
270-280	275	20		
280-290	285	10		
		$n = \sum f_i = 100$		

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6 Skewness and Kurtosis

7 Correlation analysis

8 Simple Linear Regression

9 Probability and Random variables

10 Discrete random variable and Expectation

11 Continuous random variable

12 Statistical Quality Control (SQC)

13 Summary

In summary, this book has no content whatsoever.

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References